

Modern coexistence theory

Learning objectives

1. Know the what niche and fitness differences are and how they relate to lambda and alpha
2. Understand the storage effect

last lecture: processes that lead to coexistence and diverse communities, general

today, specifically talking about modern coexistence theory

Modern coexistence theory

last week, use simulations to get invasion growth rate, which tells us about coexistence

but we don't know why a species can or can't invade

goal: determine why species can/can't coexist

1. invasion analysis
 2. partition invasion growth rates into mechanisms
 3. get invasion growth rate in different scenarios (counterfactual) and compare
- example: constant environment

Partitioning

- niche differences: species differences that make species limit itself more than others *processes we talked about last week*
- fitness differences: species differences that determine outcome without niche differences *superior vs. inferior competitor in constant env*

$$r = FD + ND$$

- weak invader -> FD term is negative *fecundity of invader is lower than average fecundity of other species*
- coexistence (+r): niche differences > fitness differences
 - species A and B prefer different environments
 - species A has average fitness advantage: 10% more seeds germinate on average
 - species B high germination in preferred environment
 - intra > inter, ND large
- compare to counterfactuals -> strength of mechanisms *somewhat similar to an elasticity analysis*

Connection to Beverton-Holt

- ND: $\alpha_{ii}, \alpha_{ij}, \alpha_{jj}, \alpha_{ji}$
- FD: α, λ, s, g *if have a seed bank*
- changes in α 's will affect ND & FD
- changes in fecundity will affect FD

More partitioning: storage effect

- environmental fluctuations
- both species store seeds in seed bank
- escape competition in bad years

draw seeds in seed bank, A years, many species A, few species B

B years, many species B, few species A

- advantage when rare

A years, A rare, really big, few species B

B years, B rare, really big, few species A

quick reference equations

$$\frac{N_{i,t+1}}{N_{i,t}} = (1 - g_i)s_i + F_i$$

$$F_i = \frac{g_i \lambda_i}{1 + \alpha_{ii} g_i N_{i,t} + \alpha_{ij} g_j N_{j,t}}$$

$$\rho = \sqrt{\frac{\alpha_{ij}}{\alpha_{jj}} \frac{\alpha_{ji}}{\alpha_{ii}}}$$

$$\frac{K_j}{K_i} = \left(\frac{\eta_j - 1}{\eta_i - 1} \right) \sqrt{\frac{\alpha_{ij}}{\alpha_{jj}} \frac{\alpha_{ji}}{\alpha_{ii}}}$$

$$\eta_i = \frac{g_i \lambda_i}{1 - (1 - g_i)(s_i)}$$